

Astronaut Lander

(Coach Scheduled Event - 45 minutes)



Description: Teams will design and construct a landing device that safely protects a simulated astronaut during a landing from a designated height. Before construction, teams will participate in a mock purchasing process, being scored on their use of a limited budget to select the materials used. Teams will also complete a written assessment covering the forces, energy transformations, and engineering principles involved in protecting astronauts during reentry and landing.

Participants per assigned Team Number: 2

- If your school has 1 team you will send 2 students
- If your school has 2 teams you will send 2 students per team number; team numbers may not intermix
- If your school has 3 teams you will send 2 students per team number; team numbers may not intermix

Safety Equipment - Teams Must Bring:

- **Safety glasses labeled ANSI Z87+ (impact rated)**
- **All competitors must wear their eye protection during the entire event.**
- **If a team does not have the required eye protection, they will be given the opportunity to obtain it, time allowing, but will not receive extra time.**
- **If a team is unable to obtain eye protection, the team will not compete and will receive a no-show score.**

Impact Safety Glasses Required



Eyeglasses are not safety rated!

Teams must bring:

- Team's "Astronaut Lander Materials Shopping List" with the cost of materials calculated (The Shopping List is a separate page printed in these rules.)
- Writing utensils
- Scissors

Teams may bring:

- 1 four-function calculator

Materials Provided at Event:

All teams will "purchase" materials from the "Store" in the Event Room to build their lander - teams must bring The Shopping List filled out

"Cost" at the "Store" is listed in parenthesis before the item.

- (Free, must be used) 1 - piece of cardboard, approximately 10 x 13 cm (4 in x 5 in)
- (Free, must be used) 1 - 147 ml (5 oz paper cup) the capsule
- (Free, must be used) 1- 2.54 cm diameter (1 in) Bouncy Ball
- (\$5 each) 3 - index cards 8 x 13 cm (3 in x 5 in)
- (\$0.50 per cm) Up to 100 cm (39 in) of string
- (\$5 each) 3 - rubber bands, choose from length sizes #64, #33, #19, sizes may be mixed
- (\$5 each) 4 - 20 cm (8 in) paper straws
- (\$0.50 per cm) Up to 100 cm (39 in) of 2.54 cm (1 in) masking tape
- (\$10 each) 10 - mini marshmallows 6 cm diameter x 9 cm height (0.23 in x 0.35 in)

- Student choice of coffee filters up to 4:
 - (\$5 each) Size #1 1 cup
 - (\$5 each) Size #2 2-6 cups
 - (\$10 each) Size #4 8-12 cups
 - (\$15 each) Size #6 Very large capacity

THE COMPETITION

The Written Assessment: The team will answer questions on a written assessment about the forces, energy transformations, and engineering principles involved in protecting astronauts during landing. The test will take place simultaneously during the construction of the team's lander.

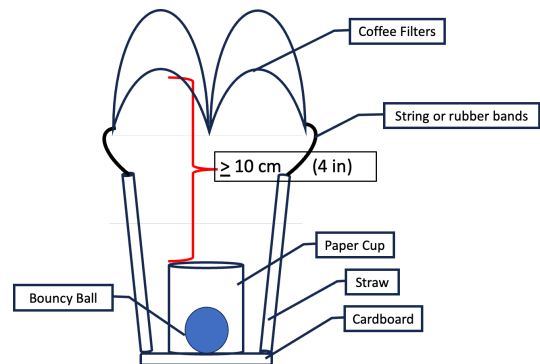
During the written test, the team will give an estimate of the flight time of their lander.

Construction Phase (25 min):

- The Event Supervisor will announce the distance for the drop to the ground floor at the start of the construction phase.
- Teams will present their Shopping List to the Event Supervisor. If the list is not present or not filled out in its entirety, the team will be allowed to create a list, but will receive a penalty to their score. The team will not receive any extra build time or ES assistance if the list has to be modified or created.
- Upon receiving and evaluating the list, the Event Supervisor will provide the “purchased” materials.
- The team will build an astronaut lander using the “purchased” materials. They do not have to use all of the materials, but they must use the paper cup, cardboard piece, and bouncy ball.
- The team may cut the materials, **except the bouncy ball, and the paper cup**, to any size needed or leave them whole
- The team may not create a method or use tape to keep the bouncy ball inside the cup and below the cup lip.
- The team may not build any part of their device within 10 cm directly over the top of the cup lip (see the diagram in the Possible Resources section).

Competition Diagram:

- This diagram is for rules understanding only. It is not meant to be a blueprint for a team design!



Launch Phase:

- The team will place the bouncy ball in the provided cup. The bouncy ball may not be wedged or taped into the provided cup
- The team will hold their lander so that the bottom of the lander is parallel to the bottom of the launch zone and announce to the Event Supervisor they are ready to drop.
- After receiving a reply from the supervisor, the team will then use a 4-point count down, such as “3, 2, 1, drop”, to start the lander falling
 - The team may not move the lander upwards or downwards while they are letting go.
- The Event Supervisor will measure and record the time for the lander to reach the ground.
- Time will continue if the lander bounces off an object but will stop when the lander gets stuck or comes to a complete stop.
- Time will be recorded in seconds to the nearest 1/100.
- The launch is considered “successful” if the bouncy ball astronaut does not exit the capsule when landing.

SCORING:

Final Score (FS) = Assessment Score + (Time Score + Shop Score) x Penalties, max 100 pts

- Assessment Score = $(TAS \div CAS) \times 40$; rounded to the 100th place value; maximum 50 points
 - Team Assessment Score (TAS) = team's score on the assessment
 - Competition Assessment Score (CAS) = the highest assessment score for all teams
- Time Score = $(TT \div CT) \times 30$; rounded to the 100th place value; maximum 50 points
 - Competition Time (CT) = the longest drop time for all teams
 - Team Time (TT) = the team's drop time
- Shop Score = $(CC \div TC) \times 30$; rounded to the 100th place value; maximum 50 points
 - Competition Cost (CC) = the lowest spent at the shop over all teams
 - Team Cost (TC) = the team's amount spent at the shop on materials
- Penalties:
 - Construction Penalty (0.8 multiplier) : A team lander that does not meet the construction guidelines or their astronaut falls out of the cup will receive a 20% penalty to their time+shop score.
 - Missing Shopping List Penalty (0.7 multiplier): A team who does not bring a completed shopping list will receive a 30% penalty to their time+shop score.
- Example
 - Competition Assessment (CAS) = 15. The team scores (TAS) 9 on the assessment. Assessment Score for the team = $(9 \div 15) \times 40 = 24$ points
 - Competition Time (CT) = 3.56 seconds. The team's flight time (TT) is 1.45. Time Score for the team = $(1.45 \div 3.56) \times 30 = 12.22$ points
 - Competition Cost (CC) = \$120, The team's cost (TC) is \$160.
 - Shop Score for the team = $(120 \div 160) \times 30 = 22.50$ points
 - Penalties - the team had an incomplete shopping list = 0.7 multiplier
 - Final Score (FS) = $24 + .7 \times (12.22 + 22.50) = 48.30$ points

Tiebreakers:

- 1st - The smallest difference between the actual and estimated flight times
- 2nd - Correct answers on selected written assessment questions

Resources:

- [Make an Astronaut Lander – Engineering Project | NASA JPL Education](#)
- Science Teachers & Librarians
- Get permission from your parents/coach to use AI to create a study guide using the rules provided here

“Purchasing” of Materials

Each team must bring to the competition their completed shopping list using the template listed below. For their shopping, each team may list any items from the list up to a total cost \$200.

Team # _____ School Name _____ Team Name _____

Astronaut Lander Materials List

Materials	Part “Price”	Max Allowed	Number Needed	Purchase Cost
Cardboard	Free	1	1	0.00
Capsule	Free	1	1	0.00
Bouncy Ball	Free	1	1	0.00
Index cards	\$5 each	3		
String	\$0.50 per cm	100 cm		
Rubber bands	\$5 each	3		
Paper straws	\$5 each	4		
Masking tape	\$0.50 per cm	100 cm		
Mini marshmallow	\$10 each	10		
Student choice of coffee filters - max 4 coffee filters total				
Size #1 1 cup	\$5 each	4 filters max over all sizes		
Size #2 2-6 cups	\$5 each	4 filters max over all sizes		
Size #4 8-12 cups	\$10 each	4 filters max over all sizes		
Size #6 Very large capacity	\$15 each	4 filters max over all sizes		
TOTAL COST				

Allowed Calculator

This is the acceptable calculators for all Division A events.

Class I - Stand-alone non-graphing, non-programmable, non-scientific 4-function or 5-function calculators. These are the most basic types of calculators. They are limited to the four basic mathematical functions and square roots. They are often cheap, costing as low as a dollar online.



Safety Glasses

Indirect Vent Goggles or Safety glasses are a mandatory safety requirement for some events. It is important to choose the type of protection specific to the task. Competitors are responsible for providing their own protective eyewear. Teams will not be allowed to compete without adequate eye protection.

Category C: Indirect Vent Goggles



Category B: Safety glasses



Compliant Eyewear Categories:

CATEGORY B

- Description: Impact protection. They provide protection from a high inertia particle hazard (high mass or velocity)
- Corresponding ANSI designation/required marking: Z87+.
- Category B eyewear can look either like goggles or safety glasses, but must have the Z87+ marking embossed on them.

CATEGORY C

- Description: Indirect vent chemical/splash protection goggles. These seal tightly to the face completely around the eyes and have indirect vents constructed so that liquids do not have a direct path into the eye (or no vents at all). If you can see through the vent holes from one side to the other, they are NOT indirect vents
- Corresponding ANSI designation/required marking: Z87 (followed by D3 is the most modern designation but, it is not a requirement)

Examples of Non-Compliant Eyewear:

- Face shields/visors are secondary protective devices and are not approved.
- Prescription Glasses containing safety glass should not be confused with safety spectacles. "Safety glass" indicates the glass is made to minimize shattering when it breaks. Unless these glasses bear the Z87+ mark they are not approved for use.

Notes:

1. A goggle that bears the Z87+ mark and is an indirect vent chemical/splash protection goggle will qualify for both Category B & C
2. VisorGogs do not seal completely to the face, but are acceptable as Type C goggles